FINDING DINO: ENHANCING ACTION POLICY PERFORMANCE UNDER LIGHTING AND CAMERA DISTORTIONS

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ABSTRACT

In this project, we propose an image processing pipeline to improve the robustness of vision-based robotic manipulation policies under varying visual conditions. We train a policy to perform a simple interaction task—touching a toy dinosaur with the robot's end effector. While the learned policy performs reliably under nominal lighting and camera settings, we observe significant degradation when exposed to out-of-distribution (OOD) conditions such as dim lighting, glare, or partial occlusions caused by mesh or cloudy films on the camera lens. These findings highlight the sensitivity of visuomotor policies to environmental variations. To address this limitation, we propose incorporating *digital image processing* techniques improve policy generalization. Our experiments aim to quantify the improvement in policy performance and visual robustness achieved through these preprocessing strategies.

***Index Terms*—** Robot Manipulation, Digital image processing, ACT

1. INTRODUCTION

Learning-based visuomotor policies have shown remarkable capabilities in enabling robots to perform manipulation tasks directly from visual inputs. However, such policies are often sensitive to changes in lighting, camera quality, and environmental appearance, which limits their generalization to real-world conditions. In this work, we study this limitation through a simple yet illustrative task using the `LeRobot` framework [1].

We trained an *Action Chunking Transformer (ACT)* policy [2] for 50 episodes to perform a basic interaction task—touching a toy dinosaur with the robot's end effector. During training, the robot operated under consistent and well-lit visual conditions. While the trained policy achieved high success rates in this setup, its performance degraded drastically when the environment lighting was altered. In particular, switching off the ambient lights—while maintaining sufficient visibility for a human observer—caused the policy to fail consistently in locating and touching the target object. The robot's actions became erratic, often missing the object entirely or stopping prematurely. A demonstration of these failed trials can be found in our supplementary video [1].

These observations highlight a critical limitation of vision-based policies trained under narrow visual distributions: even subtle deviations in illumination or camera clarity can lead to severe degradation in task performance. To address this challenge, we propose incorporating *digital image processing (DIP)* techniques to enhance visual robustness before feeding images into the policy network. In particular, we explore methods such as contrast normalization, adaptive histogram equalization, and denoising filters to compensate for lighting and lens distortions. Our ongoing experiments aim to evaluate whether such preprocessing can mitigate out-of-distribution (OOD) failures and improve the generalization of visuomotor control policies.

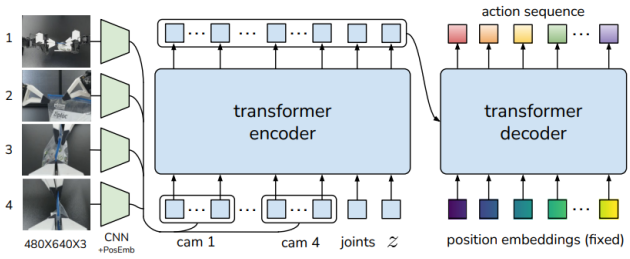


Fig. 1: Overview of the Action Chunking Transformer (ACT) architecture used as the visuomotor policy. The model receives multi-view image inputs and predicts temporally consistent end-effector actions.

2. RELATED WORK

2.1. Visuomotor Policy Learning

End-to-end visuomotor policy learning has become a powerful approach for robotic manipulation, allowing agents to map raw visual inputs directly to motor commands [3, 4]. The *Action Chunking Transformer (ACT)* [2] demonstrates that transformers can learn effective control behaviors from limited demonstrations by predicting short sequences of future actions, capturing temporal context efficiently. Diffusion-based policies frame action prediction as a conditional denoising

[1] `https://www.youtube.com/watch?v=SZPuDifZKKU`

process, enabling the modeling of multimodal action distributions for more complex control tasks [5]. Model-based reinforcement learning approaches such as TD-MPC [6] combine learned dynamics models with short-horizon planning to improve sample efficiency and long-horizon reasoning. More generally, vision-language-action (VLA) models integrate visual, action, and sometimes linguistic representations to enable instruction-conditioned or multi-modal control, with compact variants like *SmolVLA* [7] and Π_0 [8] achieving fast, data-efficient manipulation. These approaches highlight the potential of end-to-end visuomotor policies for joint perception and control, while still often assuming stable and consistent visual environments.

2.2. Robustness to Visual Domain Shifts

Although visuomotor policies have shown impressive results, they often struggle when deployed in environments that differ visually from their training data. Changes in lighting, camera angle, background clutter, or sensor noise can shift the visual input distribution, leading to poor generalization and degraded performance [9, 10]. To tackle this issue, researchers have explored several strategies. Data augmentation applies transformations such as brightness changes, color jittering, or occlusions to make models more robust to visual variation [11, 12]. Domain randomization takes this a step further by randomizing textures, lighting, and viewpoints in simulation so that trained policies can transfer more effectively to the real world [9]. Other approaches use fine-tuning or domain adaptation to adjust pretrained models with a small amount of real-world data [10]. While these methods improve robustness, they often rely on additional data collection or retraining, which can be time-consuming and impractical for simple or resource-limited robotic systems.

2.3. Image Processing for Visual Robustness

A complementary line of work focuses on enhancing visual robustness through *digital image processing (DIP)* techniques applied directly to camera inputs. Preprocessing methods such as contrast enhancement, adaptive histogram equalization, denoising, and white balance correction have been shown to improve feature stability under lighting variations[13]. These techniques can be efficiently integrated into the perception pipeline without modifying the underlying policy. In our work, we adopt this approach to preprocess images before inference, aiming to mitigate failures caused by out-of-distribution visual conditions. Histogram equalization is one such technique which can be integrated in the pipeline to improve robot inference as shown in [14]. Another image processing pipeline is outlined in [15] which used bilateral filtering and contrast limited histogram equalization to improve the quality of robotic welding.

3. METHOD

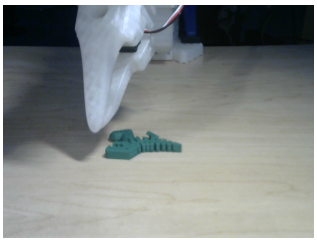
3.1. Robot Action Policy

We adopt the *Action Chunking Transformer (ACT)* [2] architecture as our base visuomotor policy for end-to-end robot control. The ACT model encodes sequences of visual observations and corresponding actions to learn temporally compressed representations of robot behavior. By attending over past states and actions, it effectively predicts the next motor command in a continuous control setting.

In our setup, the policy receives synchronized image inputs from two cameras providing complementary viewpoints of the robot workspace. These multi-view observations are processed through ResNet visual encoders and subsequently fused in the transformer backbone to produce the target end-effector action. The model is trained for 90 episodes using behavior cloning within the `LeRobot` [1] framework. The policy we trained was on 90 episodes of 10 seconds each, for 100,000 steps on a Google Colab T4 GPU for about 12 hours. The detailed configuration of the robot hardware, camera placement, and data collection protocol are provided in the *Experiments* section.

3.2. Dataset

We collect a custom dataset for training the visuomotor policy using two synchronized RGB cameras, an **egocentric end-effector camera**, mounted on the robot's wrist to provide a first-person view of the manipulation scene, and a **fixed environment camera**, positioned externally to capture a global view of the workspace.



(a) View from fixed environment camera.



(b) Egocentric end-effector camera view.

Fig. 2: Camera views for running action policy.

The clean dataset, denoted as $\mathcal{D}_{\text{clean}}$, is collected under controlled illumination using two fixed light sources — a table lamp and an overhead ceiling light. Data collection is performed during nighttime to ensure consistent lighting conditions and avoid ambient light variations. Each episode has a duration of 10 seconds, and a total of 90 episodes are recorded, equally distributed over 3 locations on the table, providing sufficient coverage of the task space.

To generate high-quality demonstrations, we employ a *teleoperation-based data collection* strategy. The robot is manually controlled by a human operator using an input device that allows intuitive manipulation of the end-effector in real time. Teleoperation enables natural and precise motion capture without requiring complex motion planning or scripted trajectories, ensuring that the demonstrations reflect realistic human-like control behavior. The teleoperated task

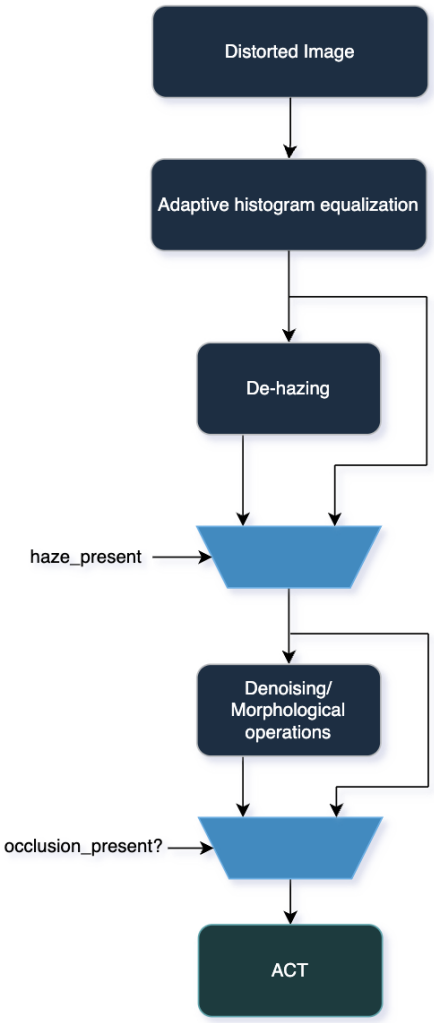


Fig. 3: Proposed image processing pipeline.

involves the robot moving its end-effector to touch a toy dinosaur placed at 3 distinct locations within a predefined workspace. The dinosaur's position is varied with some variance across trials within the field of view of both cameras to introduce spatial diversity while maintaining consistency with the robot's reachable workspace. This setup results in a well-balanced dataset suitable for learning robust visuomotor control policies.

3.3. Digital Image Processing

It is expected that the policy success rate will decrease when the input images are distorted due to non-ideal visual conditions, in contrast to the clean training distribution on which the ACT model is trained. To improve robustness under visually degraded conditions, we introduce a pre-processing stage that applies classical Digital Image Processing (DIP) techniques to the raw camera images before feeding them into the policy network. The clean dataset $\mathcal{D}_{\text{clean}}$ used for training is collected during nighttime using two fixed light sources, ensuring stable and uniform illumination. In deployment, however, the robot encounters a range of lighting and optical distortions not present in $\mathcal{D}_{\text{clean}}$, motivating the need for classical DIP corrections as a front-end module to the policy.

4. PROPOSED IMAGE DISTORTIONS AND PROCESSING ALGORITHMS

4.1. Dim Lighting (Both Light Sources off)

To simulate reduced illumination during daytime, both fixed light sources are turned off while natural ambient light is minimized. This produces uniformly dark images with reduced contrast and compressed dynamic range. The following classical enhancement algorithms are applied:

4.1.1. Gamma Correction

Gamma correction [16] performs a non-linear radiometric remapping of image intensities to selectively enhance dark regions. For a normalized intensity $I(x) \in [0, 1]$, the corrected output is given by

$$I'(x) = I(x)^{1/\gamma}, \qquad \gamma > 1.$$

Larger values of γ increase the amplification of low-intensity pixels while compressing high-intensity regions, effectively expanding shadow detail without saturating highlights. Due to its pointwise formulation, it incurs negligible computational overhead, which is important since we are running this process online for a closed loop policy.

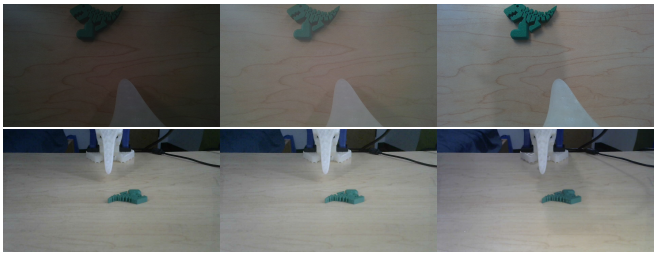


Fig. 4: The egocentric and fixed camera views showing image quality before gamma correction, after gamma correction, and the reference images captured under ideal lighting conditions.

4.1.2. CLAHE (Contrast-Limited Adaptive Histogram Equalization)

CLAHE[15] enhances local contrast by independently equalizing small spatial tiles while controlling noise amplification through histogram clipping. Given an input image partitioned into tiles, each tile computes its local histogram $h(i)$ over gray levels $i \in \{0, \ldots, 255\}$. To avoid excessive amplification of noise-dominated bins, the histogram is clipped to a predefined limit C_L:

$$h'(i) = \min(h(i),\, C_L),$$

and any excess mass is redistributed uniformly across all bins:

$$h''(i) = h'(i) + \frac{1}{L}\sum_{j}(h(j) - C_L)^{+},$$

where $L = 256$ is the number of intensity levels. A locally adaptive mapping is then generated using the cumulative distribution function (CDF):

$$T_{\text{tile}}(i) = (L-1)\,\frac{\sum_{j=0}^{i} h''(j)}{N_{\text{tile}}},$$

with N_{tile} denoting the number of pixels in the tile. To avoid block artifacts, the final output for each pixel is obtained by bilinear interpolation of the four neighboring tile mappings.

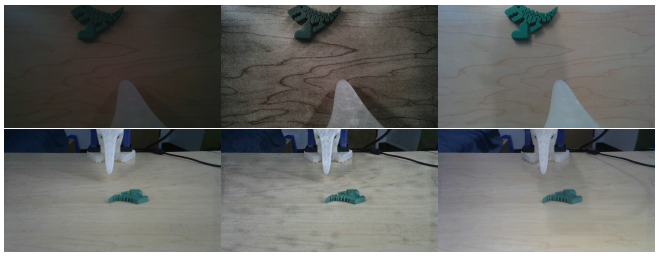


Fig. 5: The egocentric and fixed camera views showing image quality before CLAHE, after CLAHE, and the reference images captured under ideal lighting conditions.

Table 1: DIP parameters for dim lighting correction.

Processor	Parameter	Side Cam	Front Cam
CLAHE	clip_limit	1.0	5.0
	tile_grid_size	(32,32)	(4,4)
Gamma	γ	0.8	0.5

4.2. Shadow Condition (One Light Source Off)

With only one fixed light source active, cast shadows appear prominently across the workspace. Shadows introduce multiplicative illumination variation. The following classical shadow-compensation methods are applied:

4.2.1. Illumination-Blur Division (IBD)

This method models the observed image intensity as

$$I(x) = R(x)\,L(x),$$

where $R(x)$ represents the reflectance and $L(x)$ the smoothly varying illumination field. A large-kernel Gaussian blur is used to estimate $L(x)$:

$$L(x) = G_\sigma * I(x), \qquad \sigma \in [40, 80].$$

The reflectance-corrected output is obtained through pointwise division:

$$R(x) = \frac{I(x)}{L(x) + \epsilon},$$

where ϵ is a small constant to avoid division by zero. The blur removes low-frequency illumination variations, so the division step cancels multiplicative dimming caused by shadows.

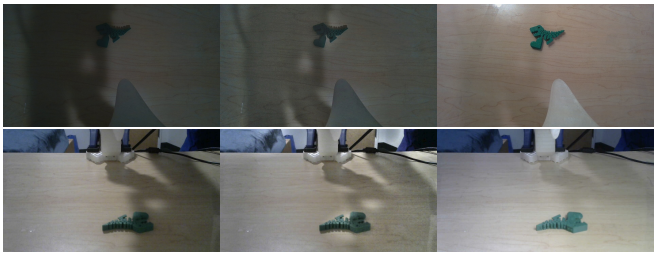


Fig. 6: The egocentric and fixed camera views showing image quality before IBD, after IBD, and the reference images captured under ideal lighting conditions.

4.2.2. Single-Scale Retinex (SSR)

SSR [17] is based on the Retinex theory, which decomposes an image into illumination and reflectance. Illumination is estimated by convolving the image with a Gaussian of scale σ, and enhancement is performed through a logarithmic ratio:

$$R(x) = \log I(x) - \log(G_\sigma * I(x))\,.$$

The logarithm emphasizes relative intensity differences while suppressing slowly varying illumination. The Gaussian scale σ controls the degree of smoothing of the illumination estimate; larger values improve shadow attenuation by removing more low-frequency variation.

Table 2: DIP parameters for shadow mitigation.

Processor	Parameter	Value
IBD	target luminance	132
	Gaussian σ	50
SSR	Gaussian σ	50
	gain	0.6



Fig. 7: The egocentric and fixed camera views showing image quality before SSR, after SSR, and the reference images captured under ideal lighting conditions.

4.3. Smudged or Dirty Lens

Lens contamination is simulated by manually introducing smudges, fingerprints, or transparent oily streaks on the camera cover. This results in spatially varying blur and contrast loss. The following classical restoration techniques are evaluated:

4.3.1. Wiener Deconvolution

Frequency-domain inverse filtering regularized by noise-to-signal ratio to recover blurred structure [18]. In the frequency domain, the wiener filter $G(f)$ is given by the formula

$$G(f) = \frac{H^*(f)S(f)}{|H(f)|^2 S(f) + N(f)}$$

$$I'(f) = G(f)\mathcal{F}(I)$$

Here $H(f)$ is the degradation function that distorted the image in the frequency domain, $S(f)$ is the power spectral density(PSD) of the original un-degraded image and $N(f)$ is the PSD of the additive noise. The final processed image is given by the inverse fourrier transform $\mathcal{F}^{-1}(I')$.

4.3.2. Richardson–Lucy Deconvolution

Iterative Poisson-based deblurring to enhance smeared details, effective for low-light sensors[19][20]. The estimated image $\hat{O}_k$ at each iteration is given by:

$$\hat{O}_k = \hat{O}_{k-1} \cdot \left[\mathbf{P}^* * \left(\frac{D}{\mathbf{P} * \hat{O}_{k-1}} \right) \right]$$

where D is the observed(degraded) image and P is the point spread function(degradation function). The number of iterations will be pre-defined such that the compute time is feasible to be implemented within the policy pipeline.

5. EXPERIMENTS

5.1. Experimental Setup

The experimental setup is designed to provide consistent and controlled conditions for both data collection and evaluation



Fig. 8: Showing the effect of the Weiner and Richardson–Lucy Deconvolution on image quality.

Table 3: DIP parameters for lens smudge and blur correction.

Processor	Parameter	Value
Wiener	kernel size	15
	Gaussian σ	3.0
	noise factor k	0.01
Richardson–Lucy	kernel size	15
	iterations	30

of the trained visuomotor policy. The robot operates within a well-defined workspace, instrumented with two cameras and two light sources, as shown in Figure 14.

The setup includes:

- The white **5 DoF Manipulator robot**, responsible for executing and following the teleoperated motions demonstrated by the pink robot, thereby reproducing the task in real time for data collection (Figure 14);
- A **fixed environment camera** mounted on a window sill, capturing the entire workspace and the robot's motion (Figure 11);
- An **egocentric end-effector camera**, attached to the robot's wrist to provide a close-up view of the toy dinosaur and interaction area (Figure 12);
- Two **lighting sources** — a table lamp and an overhead light (Figures 9 and 10) — positioned to ensure uniform illumination and minimize shadows;
- A **toy dinosaur target** placed at random locations on the table surface within the robot's reachable range defined by the transparent tape.(Figure 13).

All components are arranged to maintain consistent spatial relations throughout the experiments. The robot's base, camera mounts, and lighting positions remain fixed to prevent variations in image perspective or intensity. The use of dual light sources allows uniform brightness across the workspace while the overhead light reduces harsh shadows cast by the robot arm. Data collection and evaluation are conducted exclusively at night to avoid external illumination changes.

5.2. Proposed Experiments and Metrics

A clean dataset, denoted as $\mathcal{D}_{\text{clean}}$, is collected by manually teleoperating the white robot using the pink robot, with both

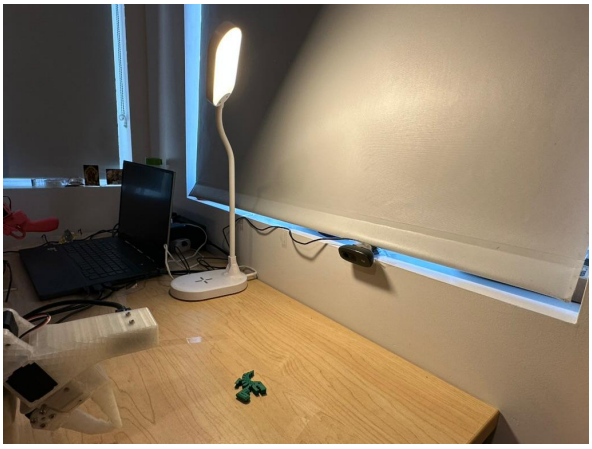


Fig. 9: Table lamp used as one of the fixed light sources to ensure consistent illumination.

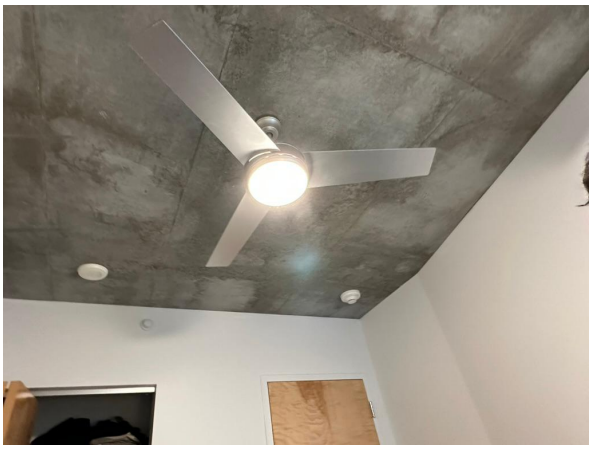


Fig. 10: Overhead light source.

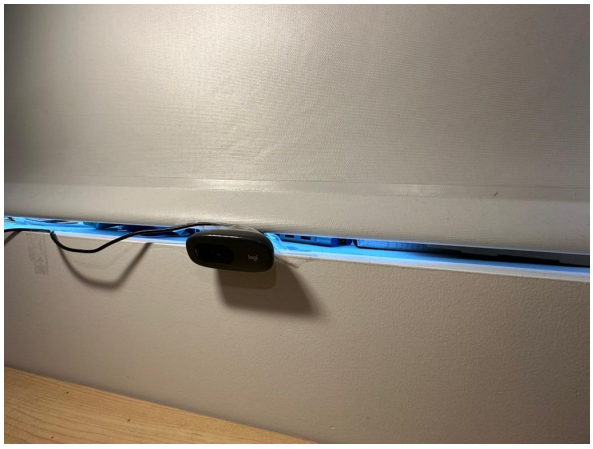


Fig. 11: Fixed external camera providing a wide-angle view of the robot workspace.

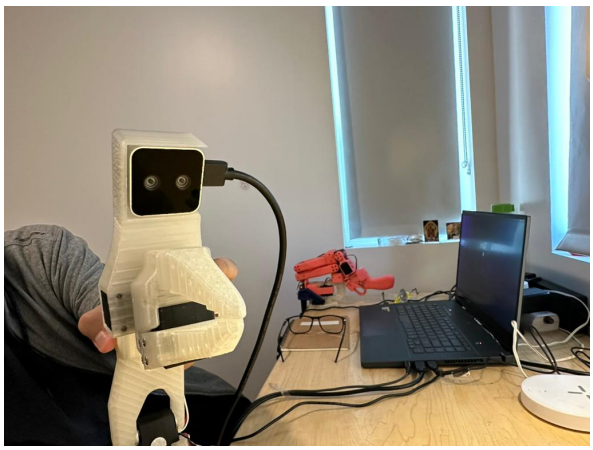


Fig. 12: Egocentric Camera on the End-effector of the robot.

light sources shown in Figures 10 and 9 turned on. The toy dinosaur is placed around three distinct positions within the robot's reachable workspace, and a total of 90 training episodes are recorded across these positions. The collected dataset is used to train the ACT policy.

For evaluation, the object is placed at the same three workspace positions used during data collection. For each position, three types of visual distortions—dim lighting, shadows, and partial occlusions—are introduced as described in Section 3.3. For each distortion type, 5 rollout trials are collected per position, resulting in a total of 15 rollouts per distortion type (3 positions $\times$ 5 trials).

All distorted images are optionally preprocessed using the proposed digital image processing (DIP) pipeline before being fed into the ACT policy trained on $\mathcal{D}_{\text{clean}}$.

The performance of the learned policy is evaluated using the **success rate**, defined as

$$S(d) = \frac{N_{\text{success}}(d)}{15},$$

where $N_{\text{success}}(d)$ denotes the number of successful rollouts under distortion type d, and 15 is the total number of trials per distortion. A rollout is considered successful if the robot correctly reaches and touches the toy dinosaur within the episode duration.

The success rate for each distortion type is compared against the baseline success rate measured under clean lighting conditions without any visual distortions.

In addition to success rate, we also evaluate the **task completion time**, defined as the elapsed time required for the robot's end effector to touch the toy dinosaur. A detailed analysis of this metric is provided in the Comparison section.

6. RESULTS

The action policy was tested with the dinosaur in 3 positions (left, right, center). The results were noted for the following three input conditions.

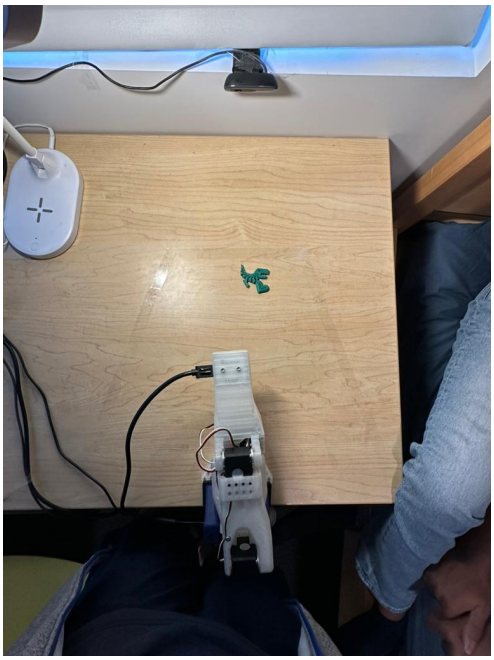


Fig. 13: Close-up of the robot's workspace showing the placement of the toy dinosaur within the reachable area. The transparent tape outlines the field of view of the fixed camera.

- Baseline: Performance of the action policy under the same lighting conditions as that of the training data.
- Distorted (Without DIP): Performance of the action policy with an out of distribution input image caused due to altered environmental conditions.
- DIP-enhanced: Performance of the action policy with the input being a distorted image upon which the image processing methods listed in sec. 4 are applied.

For each position and input condition, 5 observations were taken and the success rates and average episode times are listed below.

6.1. Shadow Distortion

Figure 15 shows results under shadow distortion. DIP-based preprocessing improves the success rate compared to the no-DIP baseline, leading to more consistent behavior and more frequent target contact. However, performance remains lower than under clean lighting, indicating that shadow-induced artifacts are only partially mitigated.

6.2. Dim Lighting

Figure 16 presents results under dim lighting. Preprocessing with CLAHE and gamma correction significantly improves success rates relative to raw inputs. Despite these



Fig. 14: Complete view of the experimental workspace showing the robots, cameras, and lighting setup. The white robot is the follower, and the pink robot is teleoperated.

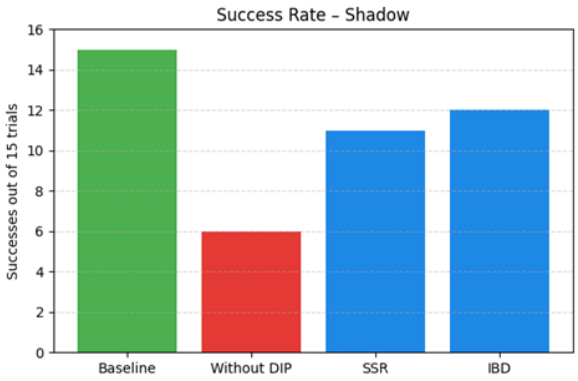


Fig. 15: Shadow

improvements, performance does not fully recover to the clean-lighting baseline.

6.3. Partial Occlusion

Figure 17 summarizes performance under partial occlusions. In this setting, DIP preprocessing provides only a marginal improvement, and in some cases the success rates with and without DIP are nearly identical, particularly for smudge-style occlusions.

This behavior is expected, as occlusions remove or corrupt image regions entirely, limiting the ability of DIP methods to recover missing visual information. As a result, unlike lighting-based distortions, the benefits of DIP preprocessing are inherently constrained in the presence of occlusions.

7. COMPARISON

Applying DIP techniques improves the success rate under dim-lighting and shadow distortions, with more marginal gains observed for partial occlusions.

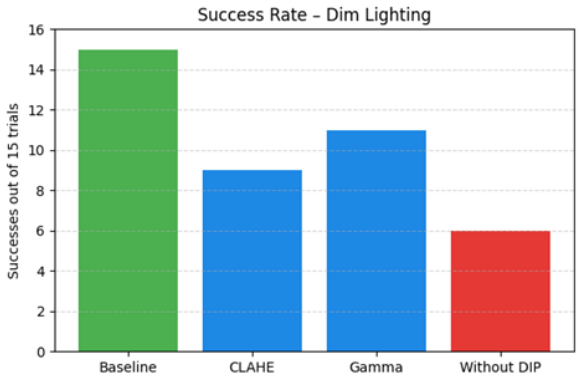


Fig. 16: Dim Lighting

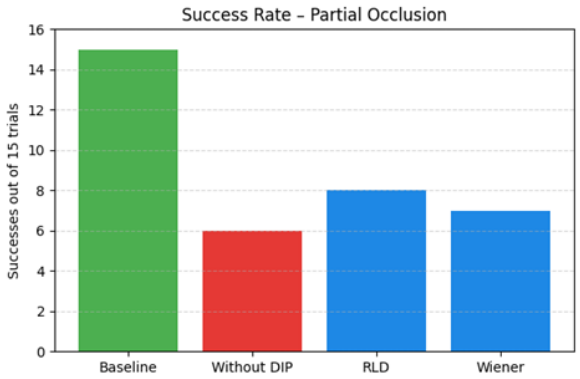


Fig. 17: Partial Occlusion

Figure 18 compares the **task completion time** for the baseline policy and various DIP-based preprocessing techniques. Performance is measured as the rollout completion time, with the mean and variance computed over fifteen trials (five rollouts for each of three target object locations).

As expected, the baseline achieves the lowest mean completion time, since evaluation conditions closely match the training distribution. All DIP methods incur a slight increase in mean completion time relative to the baseline. This overhead is primarily attributed to the additional image-processing latency introduced by DIP during closed-loop execution. Despite this increase, the variance across rollouts remains comparable, indicating that DIP preprocessing does not negatively impact execution stability.

8. DISCUSSION

An additional observation from our experiments is that target positions at the left and right extremes of the workspace result in higher error rates compared to the center position. This suggests that task difficulty increases near the workspace boundaries, likely due to reduced visibility or reachability.

Furthermore, we observe that in several cases the policy succeeds even without DIP preprocessing. We attribute this behavior to the policy being heavily conditioned on the robot's current state rather than fine-grained visual details. When the green toy dinosaur is sufficiently visible, the policy reliably infers the required direction of motion, indicating that state information plays a dominant role in action selection under certain conditions.

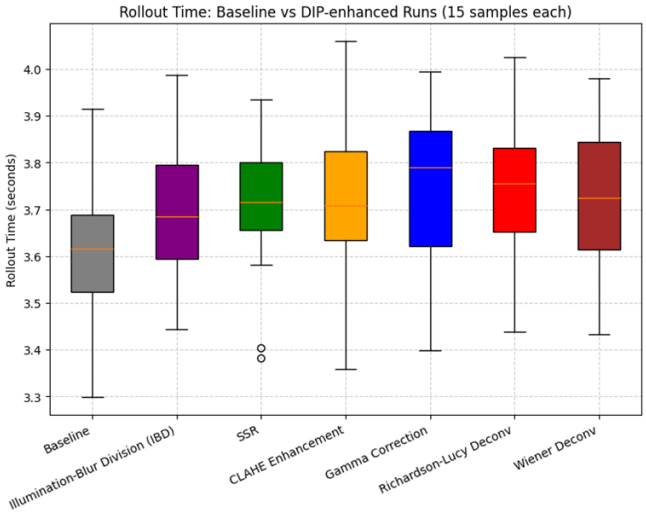


Fig. 18: Comparison of **task completion time** across baseline and DIP-based preprocessing methods.

9. FUTURE WORK

While the proposed digital image processing (DIP) pipeline demonstrates improved robustness under several visual distortions, there remain multiple directions for future investigation. First, we plan to integrate learnable perception modules, such as lightweight convolutional adapters or vision transformers, jointly trained with the action policy to complement classical DIP techniques. This hybrid approach may enable the system to adapt dynamically to unseen visual conditions.

Second, we aim to explore training-time robustness strategies, including domain randomization and data augmentation, in combination with inference-time preprocessing, to study their complementary effects on generalization.

10. CONCLUSION

We propose a simple approach to enhance the robustness of visuomotor action policies under non-ideal visual conditions. Our results show that small variations in lighting and camera quality can significantly degrade policy performance. To address this issue, we introduce a digital image processing (DIP) frontend that mitigates visual distortions before images are fed into the policy.

10.1. Effort Percentage

All team members contributed equally to this project. The effort distribution is summarized in the table below.

Table 4: Effort contribution of each team member

Team Member	Effort (%)
Umar Syed	33.3
Sachin Bhadang	33.3
Aditya Kamat	33.3

11. CODE AND POLICY ROLLOUT VIDEOS

All code and trained model weights have been open-sourced on Github [21]. The dataset used to train the policy can be accessed via the LeRobot visualization tool [22]. Finally, policy rollout videos demonstrating inference under baseline conditions as well as under various visual distortions, with and without digital image processing (DIP), are available on YouTube [23].

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